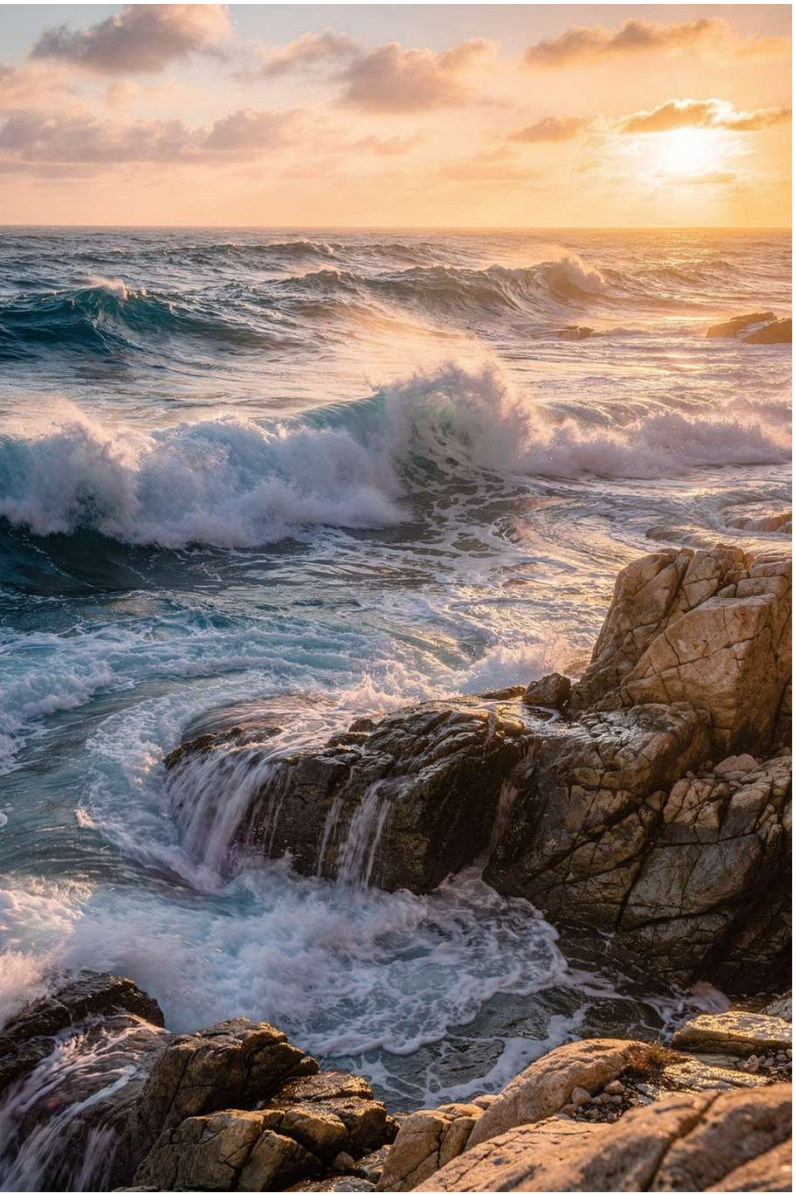


Tidal Power: Harnessing the Ocean's Energy for a Sustainable Future

Exploring how the predictable rhythm of Earth's tides can power our clean energy transition

The Celestial Dance: Moon, Sun, and Earth



Tides result from the gravitational pull of the moon and sun on Earth's oceans, creating a predictable celestial choreography. Two high tides and two low tides occur every 24 hours and 50 minutes due to Earth's rotation and the moon's orbit.

The sun's gravitational influence is approximately 46% weaker than the moon's, but their combined forces create particularly strong "spring tides" during new and full moons.

Waves vs Tides: Different Ocean Movements, Different Energy

Waves	Tides	Energy Potential
Surface oscillations caused mainly by wind transferring energy to water. Irregular in timing and magnitude, but abundant along coastlines worldwide.	***Large-scale, predictable rise and fall of sea levels driven by gravitational forces. Consistent and forecastable centuries in advance.***	***Both carry substantial kinetic and potential energy that can be converted into electricity through specialised technologies and infrastructure.***

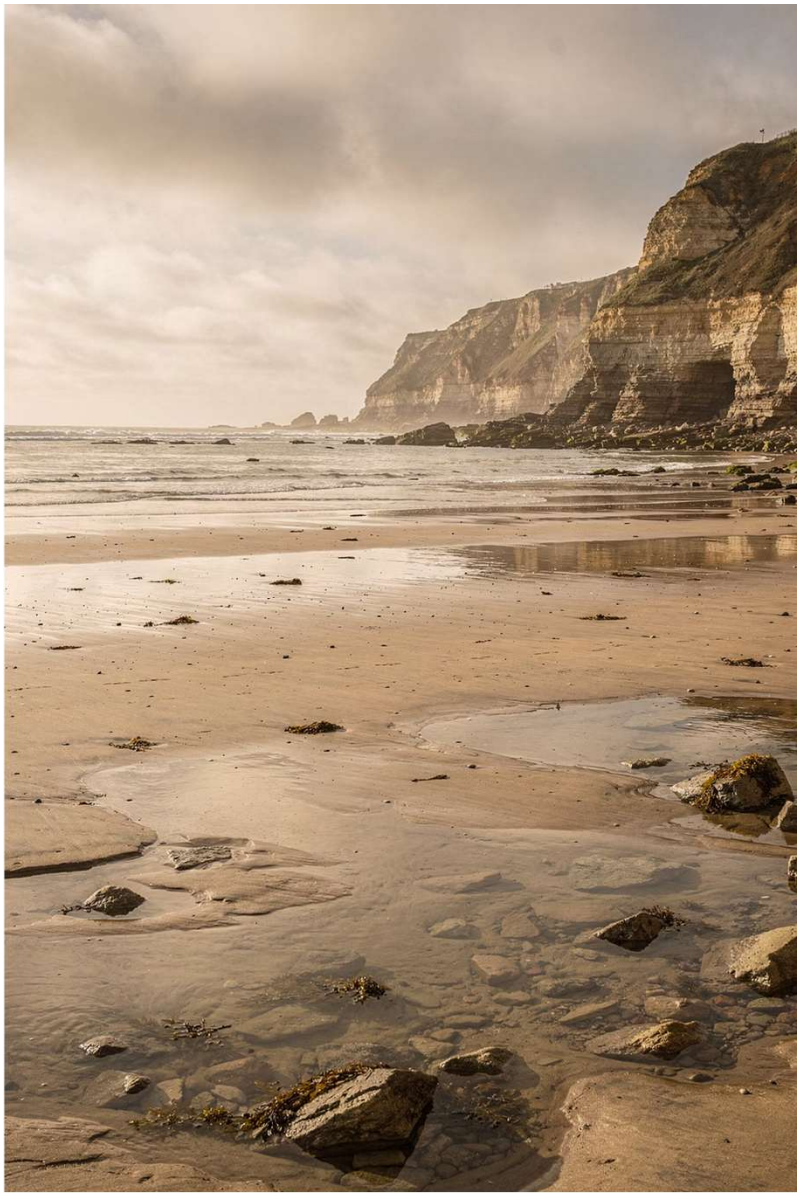
The Ocean's Pulse: What Are Tides?

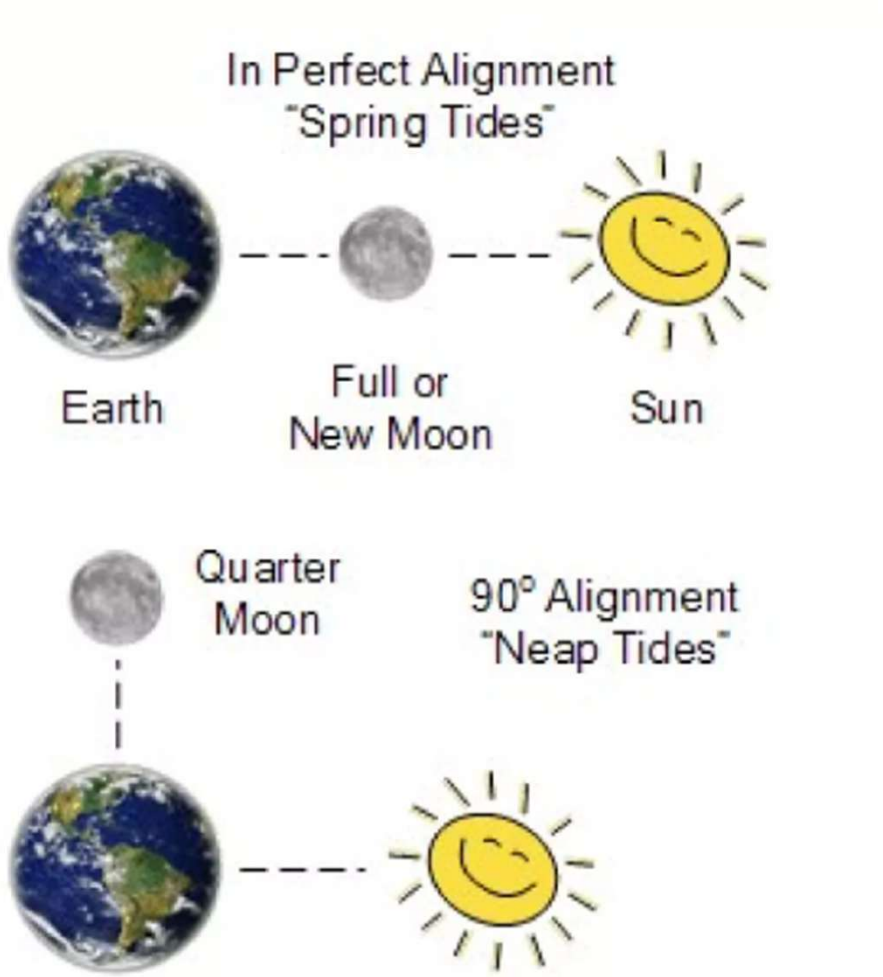
Gravitational Forces

Tides result from the gravitational pull of the Moon and Sun acting upon Earth's oceans, creating a predictable rhythm of rising and falling sea levels.

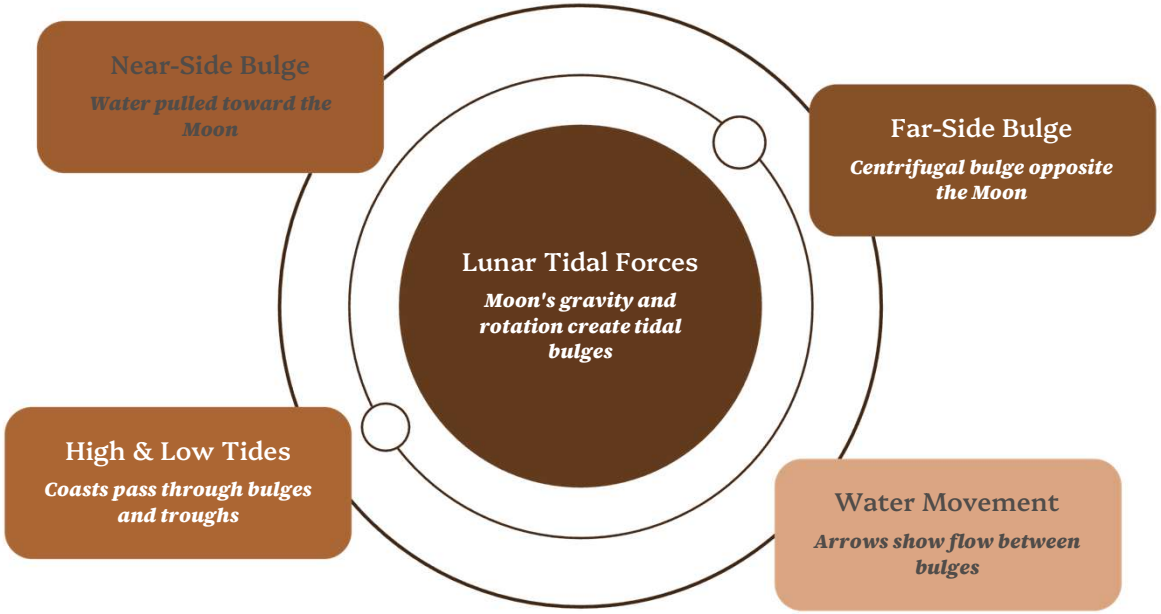
Tidal Range

The vertical difference between high tide and low tide—known as the tidal range—is the critical factor that determines a site's potential for power generation.





Alignment of the Moon and Sun on Tides



The Lunar Dance

As the Moon orbits Earth, its gravitational force pulls ocean water towards it, creating a bulge. A corresponding bulge forms on the opposite side due to centrifugal force. As Earth rotates through these bulges, coastal regions experience the rhythmic cycle of high and low tides approximately twice daily.

The Sun also influences tides, though less powerfully. When the Sun, Moon, and Earth align, we experience stronger spring tides with greater tidal ranges—ideal for power generation.

Why Tidal Energy?

A Predictable Powerhouse

Unlike wind or solar energy, which vary with weather conditions, tidal energy is extraordinarily predictable. We can forecast tidal movements years in advance with remarkable accuracy, making grid integration far more manageable.

750

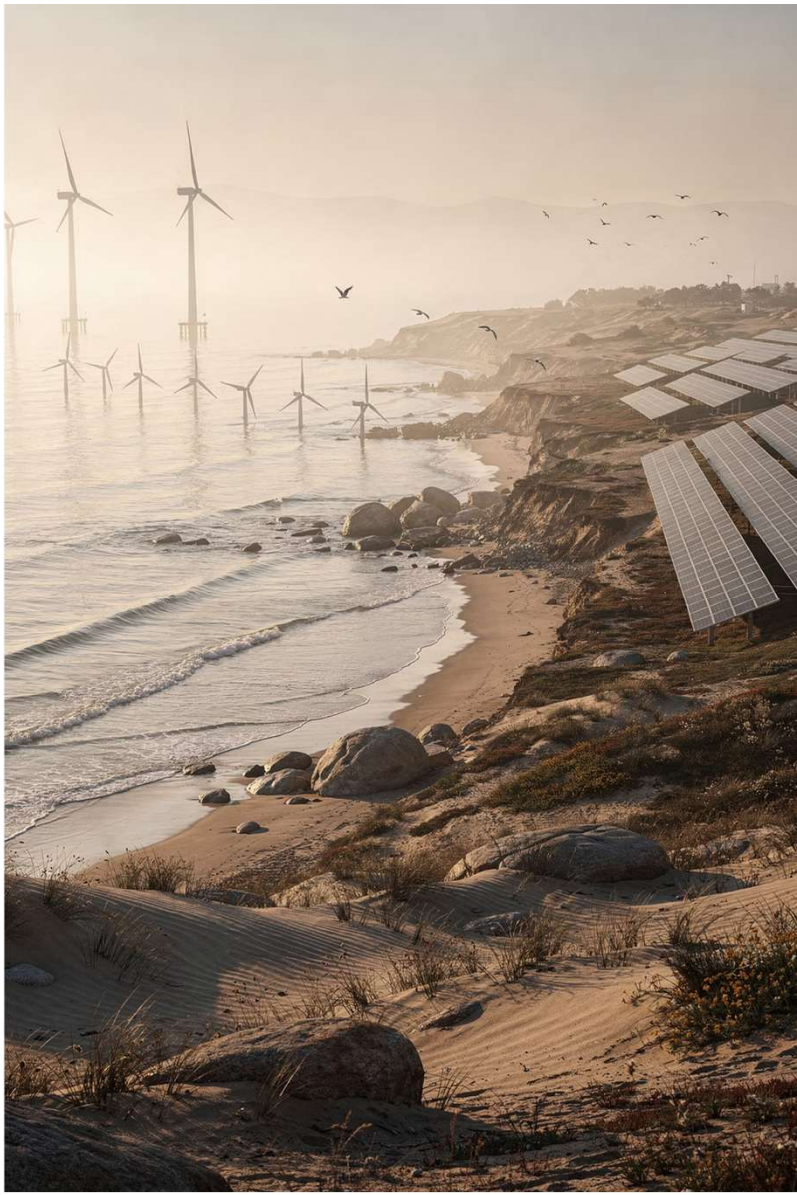
Gigawatts

Global theoretical potential of tidal energy—sufficient to power millions of homes worldwide

100%

Renewable

Zero fuel costs and no greenhouse gas emissions during operation



CHAPTER 2

Principles of Tidal Power Generation

Understanding how we transform the ocean's kinetic and potential energy into electrical power through innovative engineering and turbine technology.



Key Components of a Tidal Power Plant

Dam or Barrage

Creates a tidal basin and controls water flow between sea and reservoir.

Sluice Gates

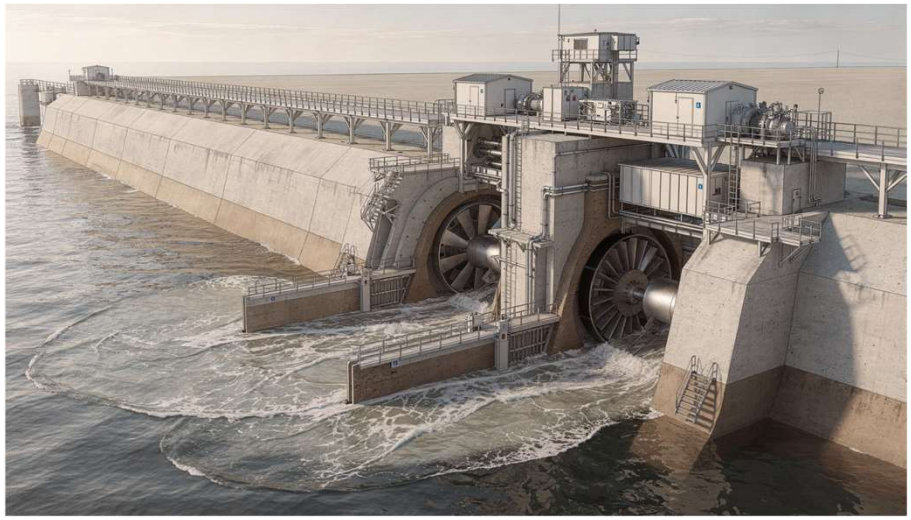
Regulate water entering and leaving the basin at optimal times.

Bi-directional Turbines

Designed to operate efficiently during both incoming and outgoing tidal flows.

Generators

Convert turbine rotation into electrical energy for the grid.



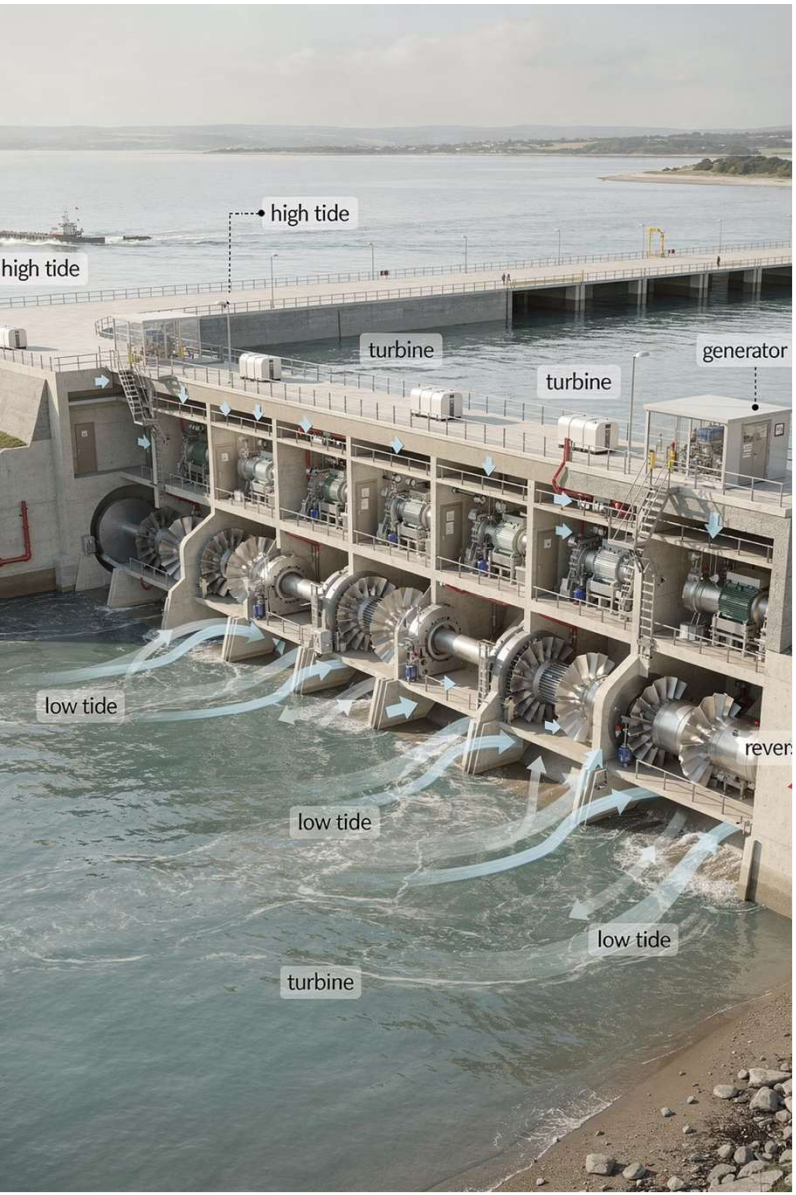


Tidal Barrages: The Classic Approach

A tidal barrage operates like a sophisticated dam built across a tidal estuary or bay. This engineering marvel traps vast quantities of seawater at high tide, then strategically releases it through turbines as the tide falls, generating substantial amounts of electricity.

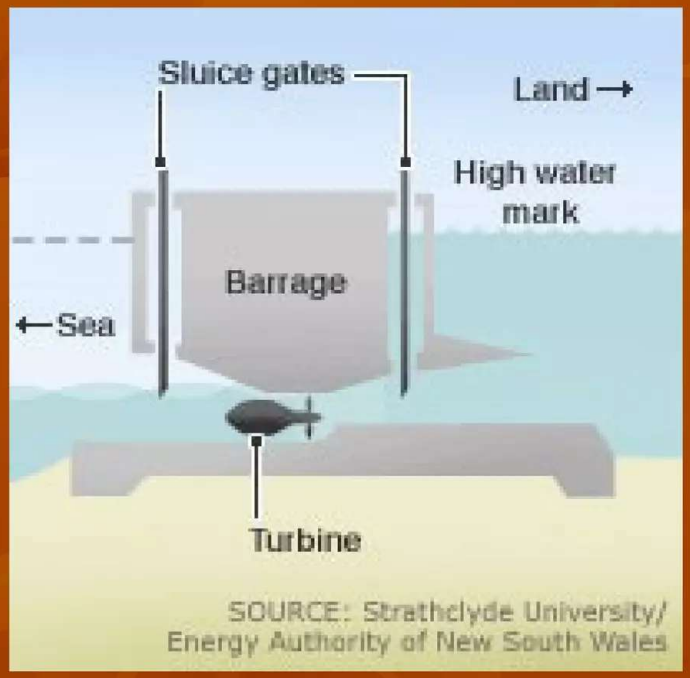
Pioneering Achievement

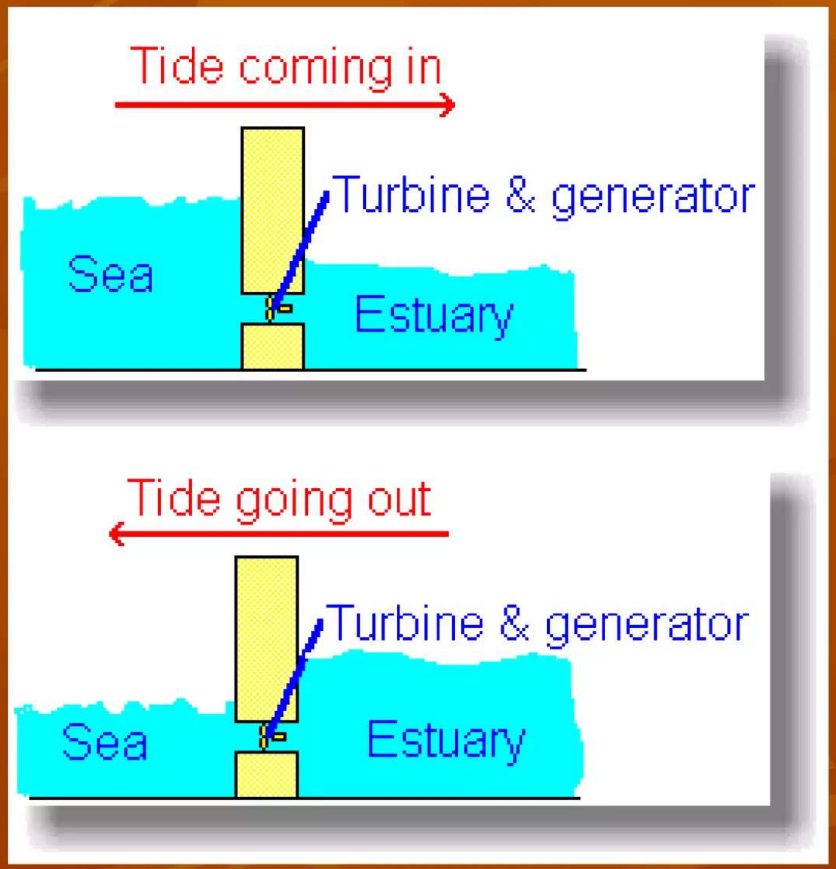
La Rance Tidal Power Station in France has been generating clean electricity since 1966, with an impressive 240 MW capacity—proving the long-term viability of this technology.



Inside a Tidal Barrage

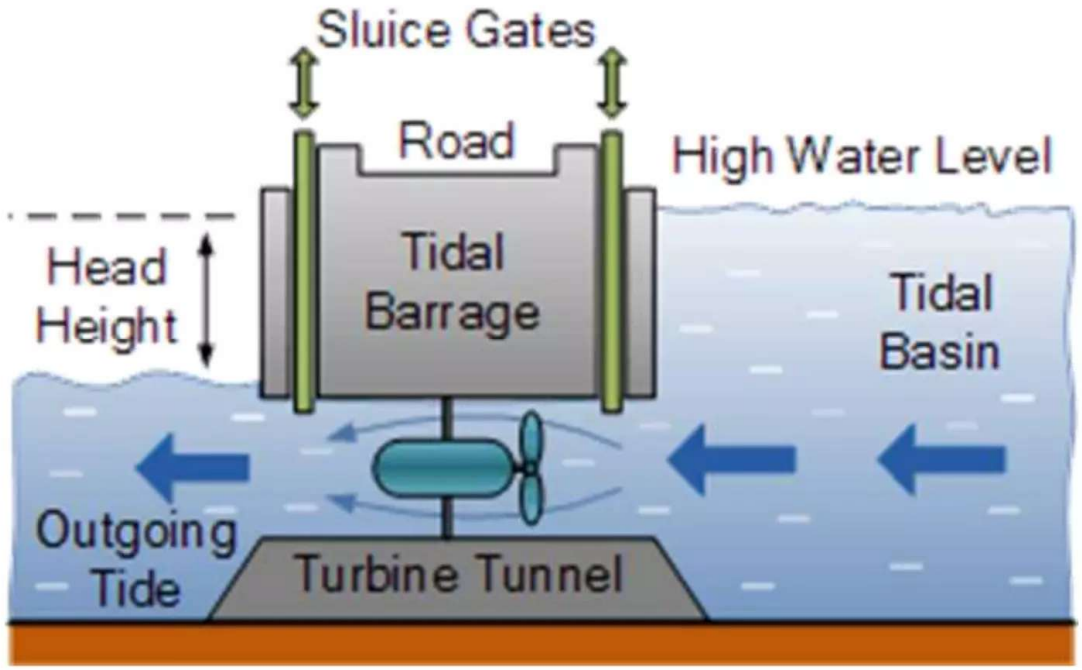
This cross-sectional view reveals how water flows through the turbine chambers during both flood and ebb tides, with sluice gates controlling the timing to maximise energy generation throughout the tidal cycle.


Sluice gates
Land →
High water mark
Barrage
←Sea
Turbine
SOURCE: Strathclyde University/
Energy Authority of New South Wales


Tide coming in
Turbine & generator
Sea
Estuary
Tide going out
Turbine & generator
Sea
Estuary

1.) Tidal Barrage

- Utilize potential energy
- Tidal barrages are typically dams built across an estuary or bay.
- consist of turbines, sluice gates, embankments, and ship locks.



SLUICE GATE



SLUICE GATE

- *A sluice is a water channel that is controlled at its head by a gate (from the dutch word "sluis"). For example, millrace is a sluice that channels water towards a water mill. The terms "sluice gate" , "knife gate", and "slide gate" are used interchangeably in thewater/wastewater control industry.*
- *A sluice gate is traditionally a wooden or metal plate that slides in grooves in the sides of the channel. Sluice gates are commonly used to control water levels and flow rates in rivers and canals.They are also used in wastewater treatment plants and to recover minerals in mining operations, and in watermills.*

OPERATION

- The term sluice gate refers to any gate that operates by allowing water to flow under it. When a sluice gate is fully lowered, water sometimes spills over the top, in which case the gate operates as a weir.
- Usually, a mechanism drives the sluice gate up or down. This may be a simple, hand-operated, worm drive or rack-and-pinion drive, or it may be electrically or hydraulically powered.

How a Tidal Barrage Works



High Tide Filling

Sluice gates open wide, allowing incoming seawater to rush into the basin behind the barrage, filling it to maximum capacity.



Water Trapped

At peak high tide, gates close securely, trapping the water and creating a substantial head difference as the ocean level begins to fall.



Power Generation

As the tide recedes, trapped water is released through turbines, spinning generators that produce electricity for the grid.



Continuous Cycle

The entire process repeats with each tidal cycle, typically generating power twice daily with remarkable consistency.

Types of Tidal Power Plants



Single-Basin One-Way

Generates power exclusively during ebb tide as water flows out. Simpler design but produces electricity only during outgoing tides.



Single-Basin Two-Way

Sophisticated systems that generate power during both flood and ebb tides, maximising energy output throughout the tidal cycle.



Tidal Lagoons

Enclosed artificial or natural lagoons that generate power as water flows in and out, offering flexibility in location and environmental impact.

Challenges in Two-Way Generation

The Engineering Challenge

Bidirectional turbines must efficiently handle powerful water flows from both directions—a significant technical hurdle requiring innovative design solutions.

Innovative Solutions

Variable Pitch Blades

Turbine blades that automatically adjust their angle to optimise efficiency regardless of flow direction.

Dual-Rotation Generators

Sophisticated generator systems designed to produce electricity efficiently with clockwise or anticlockwise rotation.

Building a Tidal Power Plant

01

Site Selection

Identify estuaries or bays with large tidal ranges (minimum ~5 metres difference between high and low tide).

02

Barrage Construction

Build robust dams or barrages spanning estuary mouths or coastal inlets using reinforced concrete and steel.

03

Turbine Installation

Install reversible turbines capable of operating efficiently in both flow directions to maximise energy capture.

04

Control Integration

Integrate sophisticated control systems for gate operation, turbine management, and grid synchronisation.

What is Tidal Power?

Core Concept

Conversion of tidal kinetic and potential energy into electricity through mechanical systems

Predictability Advantage

Unlike solar or wind, tidal energy is highly predictable and consistent, enabling accurate forecasting decades in advance.

Density Matters

Water's density is approximately 800 times greater than air, allowing high energy output even at relatively slow speeds.

Three Main Types of Tidal Power Plants



Tidal Range

Barrages and lagoons capture water at high tide behind dams, then release it through turbines at low tide, converting potential energy.



Tidal Stream

Underwater turbines positioned in fast tidal currents work like underwater windmills, harnessing kinetic energy from flowing water.



Hybrid Systems

Advanced designs combine tidal range and stream technologies or integrate wave energy capture for maximum efficiency.

Tidal Stream



Tidal streams are the horizontal flow of water, distinct from the vertical rise and fall of the tide itself.

The vertical motion of the tides near the shore causes the water to move horizontally, creating currents.

When a tidal current moves toward the land and away from the sea, it “floods.” When it moves toward the sea away from the land, it “ebbs.”

These tidal currents that ebb and flood in opposite directions are called “rectilinear” or “reversing” currents.

They can reach significant speeds, especially in constricted areas like channels or straits.

Similar to wind turbines – but more power output due to higher density of water compared to air

Real-World Examples



La Rance, France

240 MW capacity, operational since 1966. Expected lifespan exceeds 120 years, demonstrating remarkable durability.



Sihwa Lake, South Korea

254 MW capacity, the world's largest tidal barrage. Generates enough electricity to power a city of 500,000 residents.



Pentland Firth, UK

Emerging tidal stream projects harness some of the strongest currents in Europe, pioneering next-generation technology.

Harnessing the Head: Tidal Range Power Explained

Tidal barrages create a difference in water level—called "head"—between the sea and a basin. As water flows through turbines to equalise levels, mechanical energy converts to electricity.





CHAPTER 3

Advantages of Tidal Power

Tidal power offers unique benefits that distinguish it from other renewable energy sources, making it an essential component of future energy systems.

Renewable and Predictable Energy Source

Celestial Mechanics

Tides driven by moon and sun



Long-Term Predictability

Forecastable centuries in advance

Reliable Baseload

Consistent power for grids

This predictability enables utilities to integrate tidal power seamlessly into electricity grids.

Unmatched Reliability

Tides are driven by celestial mechanics that can be calculated with precision centuries in advance, unlike weather-dependent renewables.

Tidal power provides reliable baseload electricity, stabilising energy grids and reducing dependence on fossil fuels.

> ***The UK's tidal potential is estimated at 30 GW—enough to meet approximately 12% of national electricity demand.***

High Energy Density and Efficiency

Dense Medium

Water's density enables more power generation per unit area than wind or solar installations.

1

Space Efficient

Sihwa Lake's 12.7 km² footprint delivers 254 MW—far more compact than equivalent wind farms.

3

2

Slow but Powerful

Tidal turbines generate significant power even at slow water speeds due to the medium's mass.

Environmental and Economic Benefits



Zero Emissions

No greenhouse gas emissions during operation. Lifecycle emissions comparable to wind power, significantly lower than fossil fuels.



Exceptional Longevity

Infrastructure lifespan reaches 120 years for barrages—dramatically longer than solar panels (25 years) or wind turbines (20-25 years).



Low Operating Costs

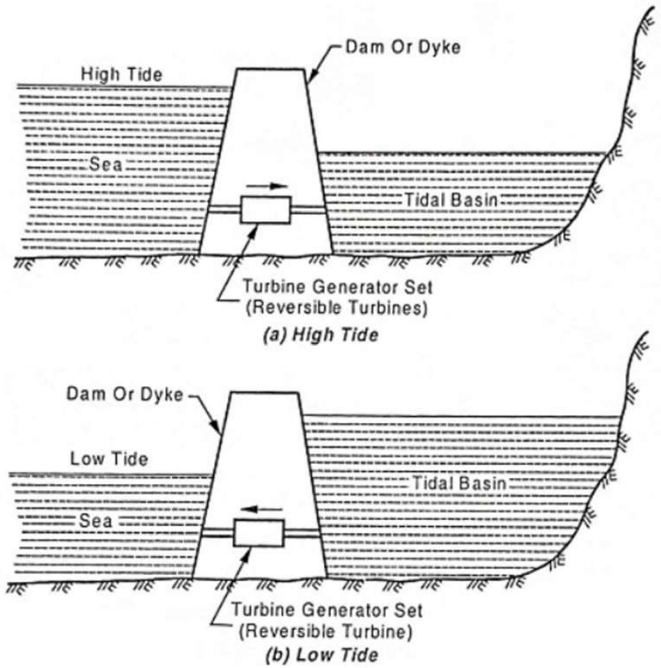
Minimal maintenance costs post-installation with no fuel requirements, ensuring stable long-term economics.

Key Classifications of Tidal Power Plants

- **Based on Number of Basins:**
  - **Single Basin Systems:** The simplest, most common type, featuring one basin separated from the sea by a dam (barrage).
  - **Double Basin Systems:** Use two basins at different levels, allowing for more continuous, controlled power generation as water is transferred between them.
- **Based on Operation Method:**
  - **Tidal Barrages:** Dams built across tidal inlets that harness potential energy by trapping water at high tide and releasing it through turbines during low tide.
  - **Tidal Stream Generators (Tidal Turbines):** Turbines placed directly in fast-moving tidal currents to capture kinetic energy, similar to underwater wind turbines.
- **Based on Cycle (Operation Timing):**
  - **Single Ebb-Cycle:** Generates power only when the tide goes out (ebbing).
  - **Single Tide-Cycle:** Generates power only when the tide comes in (flooding).
  - **Double Cycle:** Generates power on both the incoming and outgoing tides using reversible turbines.

1. Single basin arrangement

In a single basin arrangement, power can be generated only by intermittently. In this arrangement only one basin interacts with the sea. Sea and basin are separated by a dam and the flow between them is through sluice ways located conveniently along the dam. The rise and fall of tidal water levels provide the potential head.



When the incoming tide sea level and tidal-basin level are equal, the turbine way is closed.

When the sea level rises, the turbine valves are opened and the sea water flows into the basin through the turbine runner generating power.

This also raises the level of water in the basin.

The turbine continues to generate power until the tide passes through the turbine.

Bypass valve then quickly opens to let water into the basin to gain maximum level.

When sea and basin water level are equal, the valves are closed as well as the turbine conduit.

The basin level then stays constant while the tide continues to go out.

After sufficient head has developed, the turbine valves are again opened and water now flows from basin to the sea, thereby generating power.

The plant continues to generate power till the tide reaches its lowest level.

2. Double basin arrangement

A single basin plant cannot generate power continuously, though it might do so by using a pumped storage plant if the load it supplies fluctuates considerably.

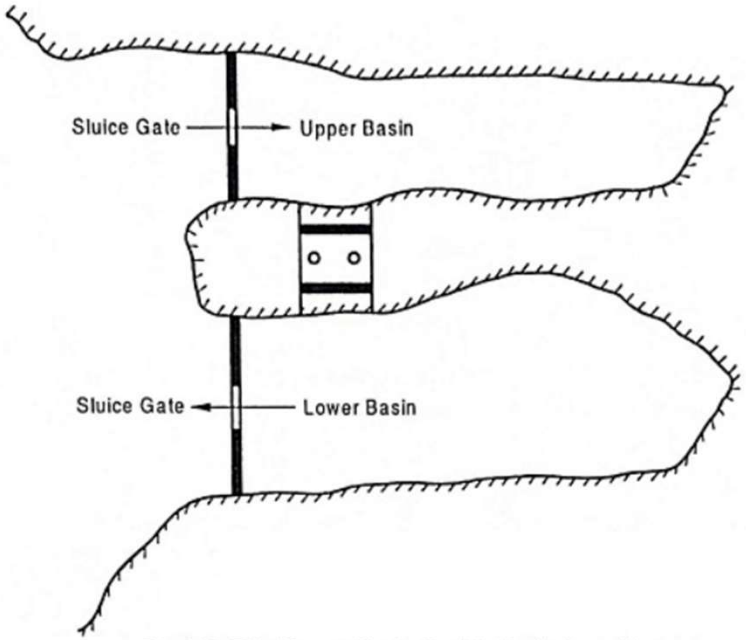


Fig:4.33. Tidal Power Plant - Double Basin Operation.

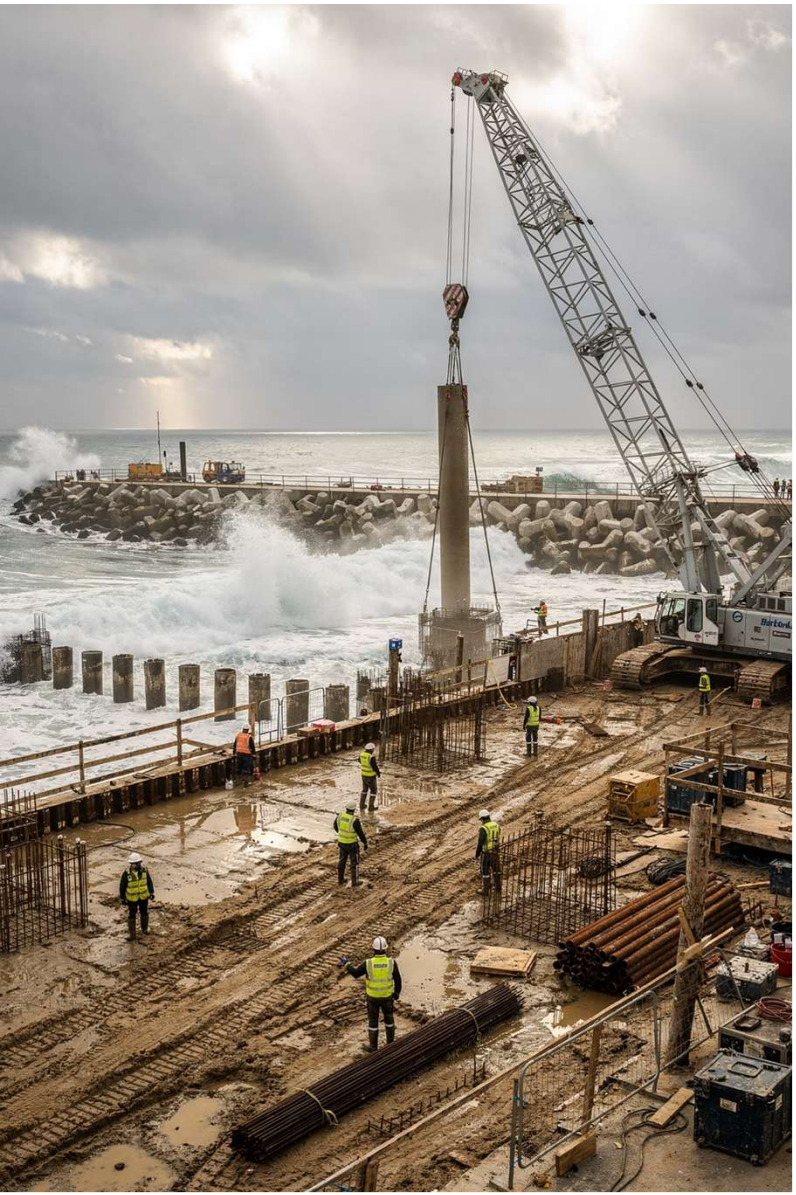
A double basin scheme can provide power continuously or on demand, which is a great advantage.

The drawbacks is that the civil works become more extensive.

In the simplest double-basin scheme there must be a dam between each basin and the sea and also a dam between the basins, containing the power house.

One basin is maintained always a lower level than the other. The lower reservoir empties at low tide, the upper reservoir is replinshed at high tide.

If the generating capacity is to be large, the reservoirs must be large which means that long dams would be required.

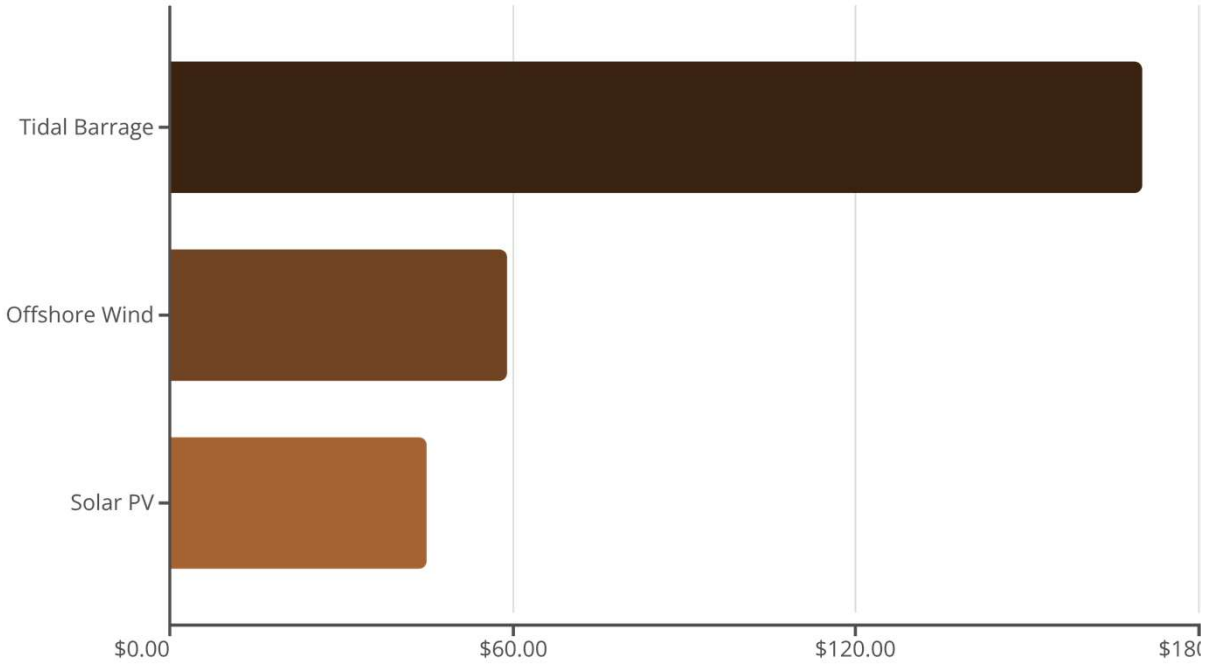


CHAPTER 4

Limitations and Challenges

Despite its advantages, tidal power faces significant hurdles that must be addressed for widespread adoption and commercial viability.

High Initial Costs and Economic Barriers



Financial Challenges

Construction costs for tidal barrage projects can reach billions of pounds upfront, creating significant financial barriers to entry.

The levelised cost of energy is currently higher than offshore wind, though costs are expected to decrease with technological advances and economies of scale.

Environmental and Ecological Concerns

Ecosystem Disruption

Barrages can significantly disrupt estuary ecosystems, alter sedimentation patterns, and affect water quality through reduced tidal flow.

Marine Life Impact

Fish migration routes may be blocked or altered. Turbine blades pose collision risks, though modern designs incorporate fish-friendly features.

Location Constraints

Viable sites limited to coastal areas with significant tidal ranges (>5 metres) or strong currents, restricting global deployment.

Maintenance Challenges

Tidal stream turbines face demanding maintenance in harsh, corrosive marine environments with strong currents and marine growth.

Environmental and Technical Challenges

Key Obstacles

Tidal power development faces significant hurdles that must be addressed through innovation and careful planning.

High Capital Costs

Substantial upfront investment due to large-scale civil engineering and specialised equipment requirements.

Marine Ecosystem Impact

Potential effects on fish migration, sediment transport, and coastal habitats require mitigation.

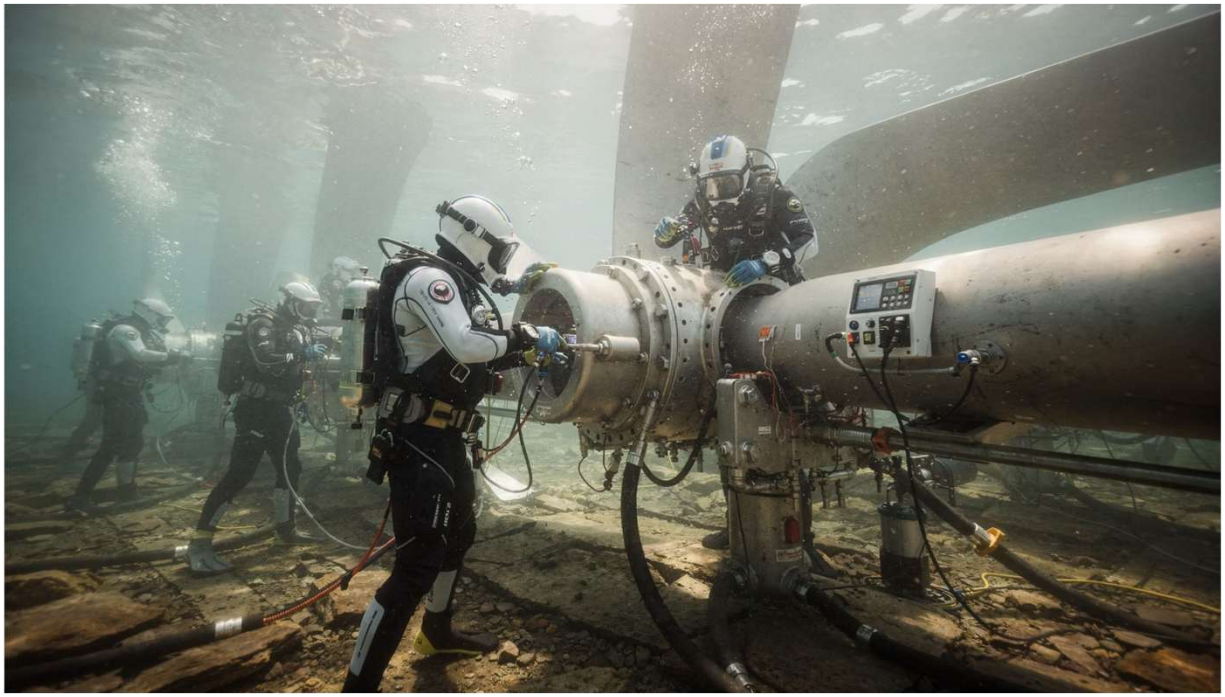
Geographic Limitations

Only feasible in select coastal regions with sufficient tidal amplitude and suitable topography.

Advanced Solutions

New turbine designs, environmental monitoring systems, and hybrid renewable approaches address concerns.

Operational Constraints



Intermittency Issues

Energy generation follows tidal cycles with short gaps approximately every six hours, requiring energy storage or complementary generation sources.

Technical Complexity

Maintenance in underwater, corrosive conditions is complex and costly, requiring specialised equipment and trained personnel.

Grid Integration

Remote coastal sites face challenges connecting to electricity grids, often requiring significant transmission infrastructure investment.

Advantages of Tidal Power Generation:

1. Tidal power is completely independent of the precipitation (rain).
2. Large area of valuable land is not required.
3. When a tidal power plant works in combination with thermal or hydroelectric system peak power demand can be effectively met with.
4. Tidal power generation is free from pollution.

Limitations of Tidal Power Generation:

1. Due to variation in tidal range the output is not uniform.
2. Since the turbines has to work on a wide range of head variation (due to variable tidal range) the plant efficiency is affected.
3. Machineries are corroded due to corrosive sea water.
4. It is difficult to carry out construction in sea.
5. As compared to other sources of energy, the tidal power plant is costly.
6. Sedimentation and silteration of basins.
7. The power transmission cost is high because the tidal power plants are located away from load centres.

Conclusion: Tidal Power's Role in a Renewable Energy Future

01

Unique Advantages

Unmatched predictability and high energy density make tidal power a valuable complement to wind and solar in diversified energy portfolios.

02

Innovation Required

Ongoing technological development and sustained investment are critical to overcoming cost barriers and environmental concerns.

03

Significant Potential

With global potential of 120 GW economically extractable capacity, tidal power can significantly contribute to decarbonisation and energy security by 2035 and beyond.

The ocean's pulse is steady—it's time we harness it wisely. By addressing current limitations through research, investment, and careful environmental stewardship, tidal power can become a cornerstone of our clean energy future.

Towards a Sustainable Ocean Energy Future

As climate challenges intensify, tidal energy's unique advantages position it as an increasingly vital component of the global renewable energy portfolio.

Technological Advancement

Continuous improvements in turbine design, materials science, and control systems are enhancing efficiency and reducing costs year on year.

Hybrid Innovation

Integration with pumped storage and tidal lagoon concepts offers unprecedented flexibility in power generation and grid management.

Global Momentum

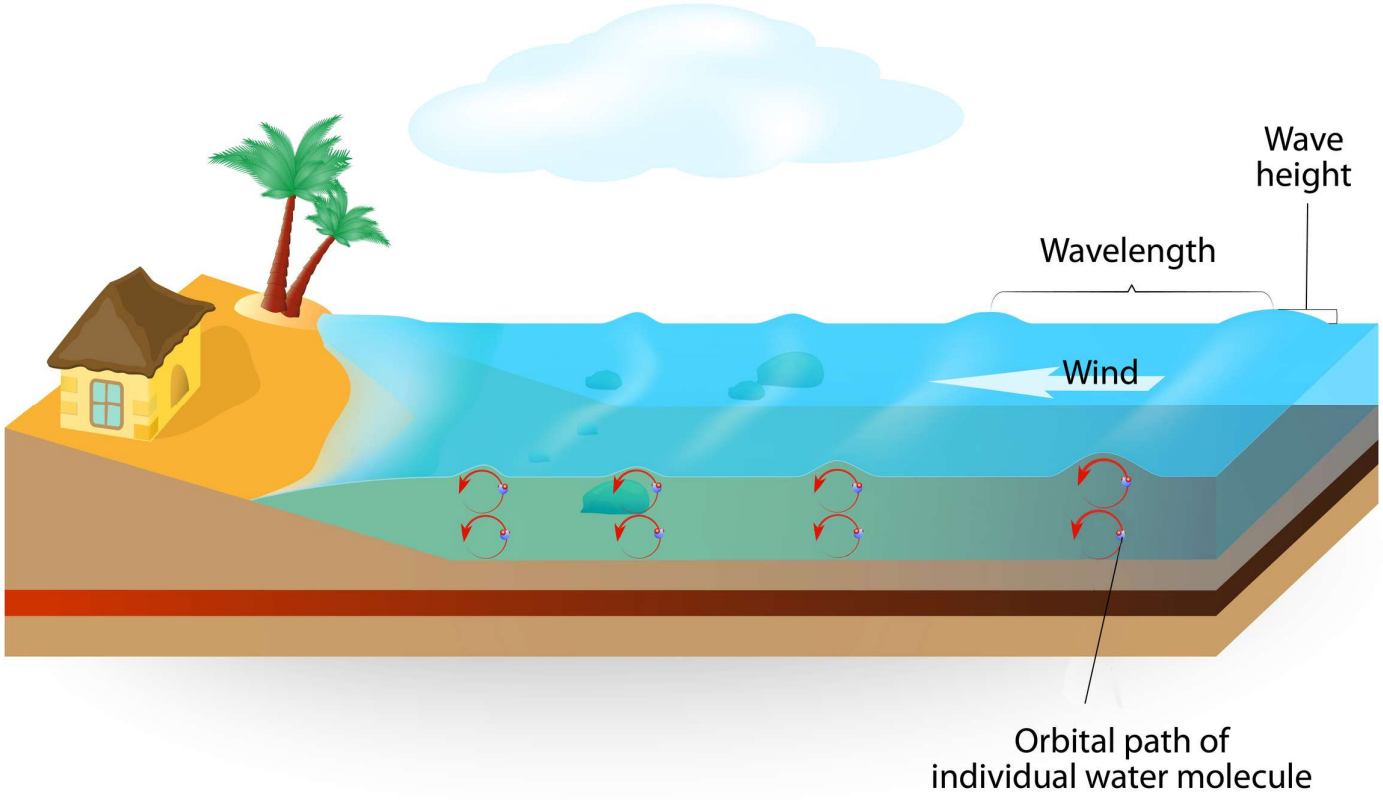
Nations worldwide are investing in tidal research and demonstration projects to diversify their renewable energy sources beyond wind and solar.

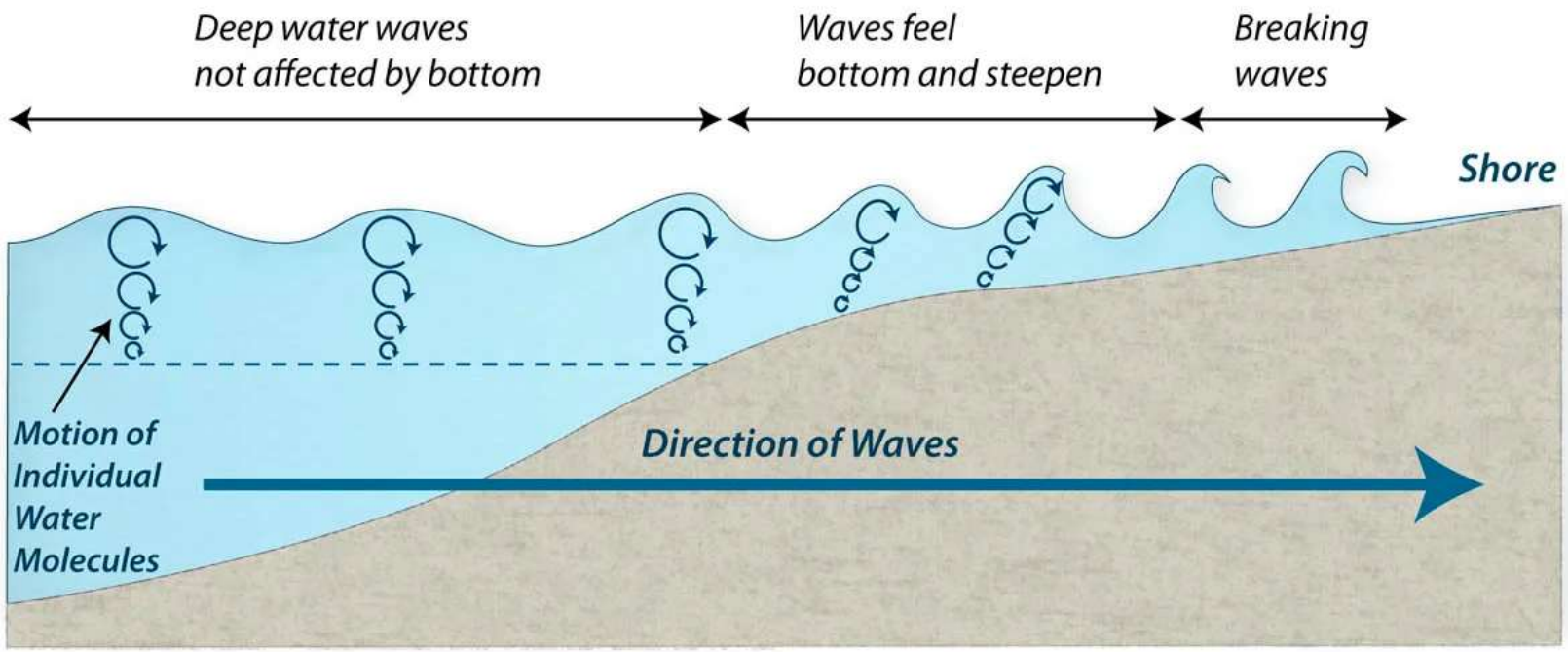
Perfect Complement

Tidal power's exceptional predictability makes it an ideal partner to variable wind and solar generation, strengthening overall grid stability and reliability.

> ***"The ocean's tides represent one of nature's most reliable rhythms. By harnessing this perpetual motion, we unlock a clean energy source that will serve humanity for generations to come."***

THE WAVE FORMATION MECHANISM




Deep water waves
not affected by bottom
Waves feel
bottom and steepen
Breaking
waves
Shore
Motion of
Individual
Water
Molecules
Direction of Waves

How Waves Form ?

- Differential warming of the earth causes pressure differences in the atmosphere, which generate winds.
- As winds move across the surface of open bodies of water, they transfer some of their energy to the water and create waves.
- A few factors determine how strong an individual wave will be. These include:
  - ✓Speed of wind: The faster the wind is traveling, the bigger a wave will be.
  - ✓Time of wind: The wave will get larger the longer the length of time the wind is hitting it.
  - ✓Distance of wind: The farther the wind travels against the wave (known as fetch), the bigger it will be.

The Power Buoy

As the buoy bobs up-and-down in the waves, a oscillatory mutual force reaction is generated between the freely moving absorber and the heavy plate causing a hydraulic pump in between to rotate a generator producing electricity.

